

Using Neural Networks to Predict Beta in Crypto and Cross-Asset Markets

Sarang Deshpande

Keywords neural networks, asset volatility, cryptocurrency

1 Introduction

The integration of cryptocurrencies into mainstream financial markets has prompted fundamental questions about how these digital assets relate to traditional investment vehicles. While classical asset pricing theory suggests that systematic risk exposures drive expected returns, recent empirical evidence indicates that cryptocurrencies may operate under a distinctly different risk-return paradigm. The relationship between cryptocurrencies and traditional safe-haven assets such as gold has proven unstable, with correlations shifting abruptly in response to macroeconomic conditions. This study extends the neural beta methodology to cryptocurrency markets, examining how Bitcoin, Ethereum, Litecoin, and Bitcoin Cash co-move with four major asset classes: fiat currencies, U.S. equities, gold, and energy. Unlike traditional linear beta estimation, neural networks can capture non-linear and time-varying relationships that may characterize crypto-market dynamics. By employing multilayer perceptrons (MLPs) with carefully tuned hyperparameters, we estimate conditional betas that adapt to evolving market conditions across different look-back windows. Our analysis addresses three core questions. First, do cryptocurrencies exhibit stable exposures to traditional asset classes, or are their betas highly volatile and regime-dependent? Second, does incorporating Fama-French factor data meaningfully improve our ability to predict cryptocurrency betas and returns? Third, is there a systematic relationship between estimated neural betas and subsequent excess returns, analogous to the risk-return trade-off observed in equity markets?

2 Implementation

2.1 Data

For this project, data on different cryptocurrencies was sourced from Coinbase via the Federal Reserve Bank of St. Louis's FRED database. The four cryptocurrencies are Bitcoin, Ethereum, Litecoin, and Bitcoin Cash. For my analysis, I also needed data on different markets, ultimately using equity, fiat, gold, and energy as my four markets. I used CRSP data from 1996-2023 to represent equity markets. As a proxy for energy markets, I used daily prices for the Vanguard Energy ETF (VDE). As a proxy for gold, I used the Credit Suisse NASDAQ Gold FLOWS103 Price Index (NASDAQQGLDI). As a proxy for fiat, I used the nominal broad U.S. Dollar index (DTWEXBGS).

The DataProcessor class handles all data loading, cleaning, and preprocessing for the neural beta prediction project. It loads cryptocurrency price data (Bitcoin, Ethereum, Litecoin, Bitcoin Cash) along with traditional market data (equity market via MSF, fiat currency index, gold, and energy sector via VDE) from CSV files in the data/ directory. The `_prepare_series` method

standardizes each dataset by converting dates, resampling to monthly frequency, forward-filling missing values, and computing percentage returns. The `_clean_asset_data` method applies business logic filters, requiring at least 3 cryptocurrencies to have valid data for each time period and forward-filling market columns. When Fama-French factors are enabled (`with_ff=True`), the class loads and merges the five-factor model data (Mkt-RF, SMB, HML, RMW, CMA, and risk-free rate). The `make_lagged_features_for_model` method creates the feature matrix by generating lagged returns for both crypto and market series up to a specified window size `w`, and optionally includes lagged Fama-French factors. The static `_month_split` method assigns each observation to train (2015-2019), validation (2020-2021), or test (2022+) sets. Tests covered: initialization with/without FF factors, the `_month_split` date classification logic, `_prepare_series` for both price and return data with forward-filling, `_clean_ff` validation filtering and RF scaling, `_clean_asset_data` market column filling and crypto availability requirements, `_merge_ff` join correctness, and `make_lagged_features_for_model` for lag creation, next-return columns, NaN handling, and row dropping.

2.2 Neural Network Architecture

The `mlp.py` file defines the neural network architecture and training infrastructure for predicting time-varying betas. The `TrainingConfig` dataclass holds hyperparameters including batch size (256), learning rate ($3e-4$), max epochs (50), weight decay ($1e-4$), and early stopping patience (5). The `NeuralBetaDataset` class wraps feature matrices and target returns into a PyTorch-compatible dataset, converting inputs to float32 tensors and validating length consistency. The `NeuralBetaMLP` class implements a simple two-layer feedforward network with configurable hidden dimension and activation function (supporting `relu`, `gelu`, `tanh`, `elu`, `sigmoid`, `identity`/linear), outputting a single scalar beta estimate. The `NeuralBetaLoss` module computes RMSE between actual crypto returns and predicted returns (`beta_hat * market_return`), using `sqrt(MSE + eps)` for numerical stability. The `NeuralBetaTrainer` class orchestrates the training loop with AdamW optimizer, ReduceLROnPlateau scheduler, and early stopping based on validation loss. The `fit` method returns the best model state, training history, and best validation loss. Tests covered: `TrainingConfig` default and custom values, `NeuralBetaDataset` initialization with arrays/lists, length mismatch detection, `__getitem__` return structure, and tensor dtype; `NeuralBetaMLP` activation function mapping including unknown activations defaulting to ReLU, custom hidden dimensions, forward pass output shapes, and determinism; `NeuralBetaLoss` epsilon handling, perfect prediction loss, gradient flow; `NeuralBetaTrainer` initialization, device placement, `train_one_epoch` and `evaluate` return types, `fit` return structure, history tracking, early stopping behavior, and `DataLoader` integration.

2.3 Fine-tuning Model Parameters

The `FineTune` class implements hyperparameter optimization via grid search over multiple dimensions to find the best neural beta model for each cryptocurrency-market pair. The constructor accepts configurable hyperparameter lists: cryptocurrencies (BTC, ETH, LTC, BCH), market combinations (equity, fiat, gold, energy), lag window sizes (12, 24, 36 months), hidden layer dimensions (4, 8, 16), learning rates ($1e-4$, $1e-3$, $1e-2$), and activation functions (linear, sigmoid, tanh, relu). The `grid_search_mlp` method iterates through all combinations, creating lagged features via `DataProcessor`, fitting a `StandardScaler` on training data, training `NeuralBetaMLP` models, and tracking the configuration with the lowest validation RMSE. It stores the best model's state dict, scaler, feature dimension, and training history. The `generate_test_predictions` method runs grid search separately for each crypto-market pair, loads the best model, and generates beta predictions on the test set, returning a long-format `DataFrame` of predictions and a metadata `DataFrame` of best configurations. The class gracefully handles missing data by skipping combinations with

insufficient training (<10) or validation (<5) samples. Tests covered: default and custom initialization parameters, `grid_search_mlp` return structure and types, best combo structure, fitted scaler verification, state dict tensor types, history structure, handling of missing cryptos, multiple cryptos/markets exploration, validation loss type and minimum tracking, feature dimension correctness; edge cases for empty DataFrames and insufficient train/val data; Fama-French factor integration increasing input dimensions; all activation functions working correctly; and grid search exploring all hyperparameter combinations with correct result counts.

2.4 Analysis

The Analyzer class provides visualization and statistical analysis tools for interpreting neural beta predictions. It maintains cryptocurrency color mappings for consistent plotting (BTC orange, ETH purple, LTC blue, BCH green) and supports both baseline and Fama-French model modes. The `plot_learning_curves` method generates train/validation RMSE plots over epochs with dynamic titles based on model type. The `compute_descriptive_statistics` method calculates comprehensive statistics for each crypto-asset pair including mean, standard deviation, skewness, excess kurtosis, and percentiles (1st through 99th), outputting to CSV when requested. The `plot_neural_beta_dynamics` method creates a 2x2 subplot showing annual mean betas with standard deviation bands for each asset class, allowing visual comparison of beta evolution across cryptocurrencies over the test period. The static `compare_baseline_vs_ff` method merges baseline and Fama-French results, computes improvement metrics, and generates side-by-side bar charts comparing validation RMSE and percentage improvements. The `create_portfolio_analysis` method constructs beta-sorted portfolios (quartiles) for each asset-month, computing both equal-weighted and value-weighted (by absolute mean return) statistics, and calculates high-minus-low (HML) spreads to assess whether neural betas predict excess returns.

2.5 Pipeline

The `main.py` file serves as the entry point that orchestrates the complete neural beta prediction pipeline. It sets reproducibility by fixing random seeds (20030910) across Python’s random, NumPy, and PyTorch, and configures pandas display options. The `main(with_ff)` function executes the full workflow: instantiates `DataProcessor` to load and preprocess data, creates a `FineTune` instance to perform grid search hyperparameter optimization, generates test set predictions using the best model for each crypto-market pair, and uses `Analyzer` to produce outputs including learning curves, descriptive statistics CSV, neural beta dynamics plots, and portfolio statistics. All outputs are saved to the `outputs/` directory with suffixes (`_ff` for Fama-French models) for differentiation. When executed directly, the script runs the pipeline twice—once without Fama-French factors (baseline) and once with them—then calls `Analyzer.compare_baseline_vs_ff` to generate comparison visualizations showing which approach performs better for each pair. The script prints the best hyperparameter configuration for each run and portfolio analysis summaries including HML spreads.

2.6 Repository

The repository for this project can be found [here](#).

3 Results and Analysis

3.1 Learning Curves

Figure 1 shows the learning curves across epochs for the baseline model. We can see that the RMSE for the training data remains constant around 0.29, while the validation RMSE is also constant

around 0.19. This indicates that the model has stopped learning and plateaued in performance, which can also be seen by the low number of epochs. Given that 0.19-0.29 is somewhat high, this could mean that the model is underfitting. To combat this, further analysis could incorporate more hidden layers or nodes, but at the cost of longer runtimes.

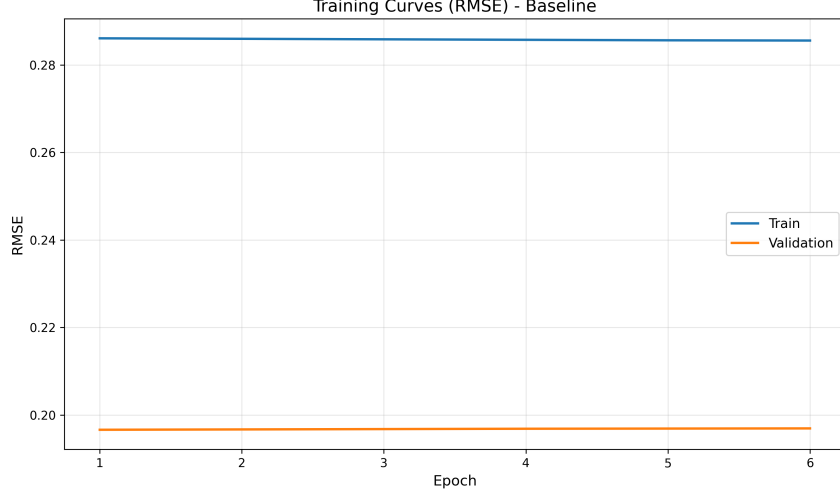


Figure 1: Learning Curves for Baseline Model

Figure 2 shows the learning curves across epochs for the model trained with Fama-French data on top of the asset data. We can see that the training RMSE starts around 0.29, but steadily falls to 0.28 over 30 epochs. In addition, the validation RMSE starts around 0.21 and falls to 0.2. This indicates that the inclusion of the Fama-French factors allowed the model to learn some relationship over the course of the 30 epochs, which marginally improved learning.

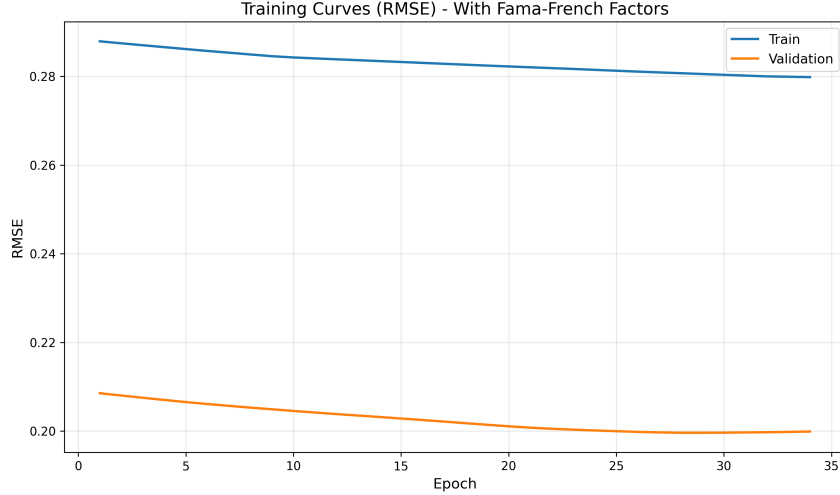


Figure 2: Learning Curves for Fama-French enhanced Model

3.2 Descriptive Statistics

Table 1 shows the summary statistics for the beta of different crypto-asset pairs based on the baseline model. The summary statistics reveal substantial heterogeneity in neural beta predictions across cryptocurrency-asset pairs, with notable differences in both central tendency and distributional properties. LTC-equity exhibits the highest mean beta exposure at 1.167 with remarkably low volatility ($SD = 0.123$), indicating stable and consistently high systematic risk relative to equity markets, while BTC-fiat displays the most negative average exposure at -0.808 with similarly

low variability ($SD = 0.079$), suggesting a persistent inverse relationship. The widest variation appears in LTC-gold ($SD = 1.374$) and ETH-energy ($SD = 1.394$), reflecting substantial time-varying dynamics in these relationships, with ETH-energy spanning an extreme range from -3.252 to 2.663. Distributional asymmetries are particularly pronounced for BTC-equity (skewness = 1.373, excess kurtosis = 2.224), indicating occasional spikes toward higher equity exposure, while BTC-gold shows strong negative skewness (-0.827), suggesting more frequent extreme negative betas than positive ones. The interquartile ranges vary considerably across pairs: BTC-fiat exhibits a compressed IQR of only 0.085 units, reflecting stable predictions throughout the test period, whereas LTC-gold spans 1.843 units between the 25th and 75th percentiles, demonstrating dramatic shifts in predicted exposures. These patterns underscore the challenge of cryptocurrency beta prediction, particularly for pairs involving traditional safe-haven assets like gold, where the relationships appear most volatile and regime-dependent.

Table 1: Summary Statistics of Neural Beta Predictions by Asset Pair

Pair	N	Mean	SD	Skew	Kurt	Min	1%	5%	25%	50%	75%	95%	99%	Max
BCH-energy	23	0.232	0.622	0.739	-0.056	-0.594	-0.587	-0.559	-0.181	0.157	0.384	1.475	1.610	1.637
BCH-equity	23	0.407	0.237	-0.666	1.227	-0.247	-0.185	0.043	0.322	0.417	0.514	0.800	0.829	0.833
BCH-fiat	23	-0.361	0.610	0.520	-0.712	-1.354	-1.292	-1.060	-0.828	-0.452	-0.027	0.689	0.855	0.895
BCH-gold	23	-0.409	0.708	0.283	-1.048	-1.534	-1.502	-1.378	-0.923	-0.495	0.135	0.696	0.843	0.880
BTC-energy	23	0.164	0.410	-0.304	-0.266	-0.782	-0.698	-0.401	-0.107	0.244	0.427	0.797	0.883	0.902
BTC-equity	23	0.840	0.471	1.373	2.224	0.292	0.301	0.335	0.510	0.814	1.002	1.503	2.139	2.315
BTC-fiat	23	-0.808	0.079	-0.393	-0.362	-0.979	-0.972	-0.943	-0.845	-0.792	-0.761	-0.683	-0.675	-0.674
BTC-gold	23	0.283	0.753	-0.827	-0.309	-1.406	-1.332	-1.068	-0.041	0.561	0.882	1.039	1.286	1.353
ETH-energy	23	0.121	1.394	-0.545	0.622	-3.252	-3.161	-2.661	-0.515	0.256	0.886	2.303	2.600	2.663
ETH-equity	23	0.427	0.684	-0.289	-0.317	-1.172	-1.017	-0.461	-0.043	0.460	0.875	1.413	1.591	1.635
ETH-fiat	23	-0.655	0.312	-0.658	-0.631	-1.312	-1.304	-1.255	-0.861	-0.613	-0.377	-0.314	-0.311	-0.310
ETH-gold	23	0.500	0.495	-0.048	-0.570	-0.474	-0.441	-0.306	0.210	0.522	0.707	1.216	1.340	1.375
LTC-energy	23	0.558	0.192	0.532	-0.926	0.318	0.319	0.326	0.401	0.503	0.668	0.838	0.931	0.957
LTC-equity	23	1.167	0.123	0.043	-0.857	0.933	0.948	1.003	1.070	1.167	1.256	1.352	1.395	1.405
LTC-fiat	23	-0.508	1.190	-0.202	-1.029	-2.635	-2.600	-2.448	-1.198	-0.109	0.274	1.130	1.230	1.259
LTC-gold	23	0.355	1.374	0.065	-0.858	-1.960	-1.935	-1.816	-0.471	0.256	1.372	2.549	2.746	2.793

Note: SD = Standard Deviation; Skew = Skewness; Kurt = Excess Kurtosis. Percentiles denoted as P1, P5, P25, P50 (median), P75, P95, P99. All statistics calculated from test period predictions (N=23 months).

Table 2 shows the summary statistics for the beta of different crypto-asset pairs based on the model with Fama-French factors. The incorporation of Fama-French factors produces markedly different distributional characteristics compared to the baseline model, with several pairs exhibiting substantially higher volatility and wider prediction ranges. Most notably, BCH-gold displays extreme variability ($SD = 2.046$) with predictions spanning from -3.851 to 5.346, suggesting the factor-augmented model captures dramatic regime shifts in this relationship that the baseline model smoothed over. ETH-fiat shows the most consistently negative exposure (mean = -1.342, median = -1.348) with relatively tight dispersion ($SD = 0.431$), indicating the Fama-French factors reinforce a stable inverse relationship between Ethereum and fiat currencies throughout the test period. In contrast to the baseline model where LTC-equity exhibited the highest and most stable beta, the factor-augmented predictions show substantially more variation ($SD = 1.201$ versus 0.123 in baseline) and a lower mean (0.492 versus 1.167), spanning from -2.149 to 3.639. The LTC-gold relationship undergoes a complete reversal, shifting from positive exposure in the baseline (mean = 0.355) to consistently negative in the Fama-French model (mean = -0.985, with 75% of predictions below -0.324), accompanied by strong negative skewness (-0.830) indicating occasional extreme negative betas. Several pairs exhibit substantial positive skewness, particularly BCH-gold (0.426) and LTC-fiat (0.275), suggesting the factor-augmented model predicts occasional spikes toward higher exposures. The generally wider standard deviations and expanded ranges across most pairs indicate that while the Fama-French factors may capture additional time-varying dynamics, they also introduce greater uncertainty into the predictions, consistent with the mixed RMSE performance observed in the comparative analysis.

Table 2: Summary Statistics of Neural Beta Predictions with Fama-French Factors by Asset Pair

Pair	N	Mean	SD	Skew	Kurt	Min	1%	5%	25%	50%	75%	95%	99%	Max
BCH-energy	23	0.172	0.514	-0.291	0.224	-1.124	-0.976	-0.441	-0.183	0.209	0.395	0.955	1.083	1.114
BCH-equity	23	0.345	0.690	0.357	-0.440	-0.728	-0.710	-0.641	-0.158	0.374	0.730	1.428	1.797	1.898
BCH-fiat	23	-0.718	1.032	-0.053	-0.811	-2.934	-2.698	-1.850	-1.530	-0.664	0.117	0.656	0.922	0.996
BCH-gold	23	0.327	2.046	0.426	0.196	-3.851	-3.416	-1.858	-1.075	0.069	1.484	3.477	4.952	5.346
BTC-energy	23	0.513	0.465	-0.049	-0.780	-0.365	-0.356	-0.291	0.185	0.518	0.803	1.194	1.235	1.246
BTC-equity	23	0.526	1.248	-0.000	-0.616	-2.020	-1.784	-0.946	-0.584	0.548	1.400	2.135	2.889	3.096
BTC-fiat	23	-0.153	0.745	0.230	-0.480	-1.521	-1.446	-1.170	-0.700	-0.171	0.274	1.242	1.311	1.316
BTC-gold	23	0.041	0.912	-0.320	-0.436	-2.032	-1.878	-1.323	-0.546	0.006	0.654	1.370	1.486	1.513
ETH-energy	23	-0.158	1.003	0.097	0.307	-2.434	-2.276	-1.635	-0.786	-0.168	0.421	1.485	1.955	2.068
ETH-equity	23	0.968	0.858	0.193	-0.183	-0.790	-0.707	-0.342	0.468	0.875	1.501	2.394	2.699	2.777
ETH-fiat	23	-1.342	0.431	0.150	-1.112	-1.951	-1.949	-1.935	-1.744	-1.348	-1.031	-0.642	-0.572	-0.557
ETH-gold	23	0.478	0.597	-0.155	-0.026	-0.893	-0.817	-0.493	0.150	0.399	0.960	1.328	1.587	1.655
LTC-energy	23	0.731	0.189	-0.227	-1.066	0.376	0.390	0.440	0.605	0.728	0.913	0.951	1.014	1.031
LTC-equity	23	0.492	1.201	0.241	0.900	-2.149	-1.913	-1.076	-0.313	0.621	1.023	2.020	3.293	3.639
LTC-fiat	23	-0.534	1.129	0.275	-0.401	-2.537	-2.474	-2.227	-1.091	-0.782	-0.012	1.509	1.696	1.735
LTC-gold	23	-0.985	0.909	-0.830	-0.285	-3.060	-2.985	-2.700	-1.545	-0.657	-0.324	0.034	0.076	0.088

Note: SD = Standard Deviation; Skew = Skewness; Kurt = Excess Kurtosis. Percentiles denoted as P1, P5, P25, P50 (median), P75, P95, P99. All statistics calculated from test period predictions with Fama-French factors (N=23 months).

3.3 Beta Dynamics

Figure 3 shows that the baseline neural network predictions reveal distinct temporal patterns in cryptocurrency beta coefficients across different asset class pairs during the 2022-2023 test period. In the crypto-energy relationship, most assets exhibit positive but modest betas, with Litecoin (LTC) showing the strongest positive exposure that increases from approximately 0.4 to 0.7 over time. Notably, Ethereum (ETH) displays negative beta initially before converging toward neutral territory. The crypto-equity panel demonstrates more stable dynamics, with all four cryptocurrencies maintaining relatively constant exposures throughout the period, though LTC consistently exhibits the highest systematic risk at approximately 1.1-1.2. Against fiat currencies, the baseline model captures a general trend of decreasing beta magnitudes, with Bitcoin (BTC) and BCH both moving from slight positive exposures toward negative territory. The crypto-gold relationship presents the most heterogeneous patterns: while LTC maintains consistently negative exposure around -1.0, BCH shows a dramatic trajectory from approximately -0.7 to +1.3, suggesting evolving correlations between this particular cryptocurrency and the traditional safe-haven asset. These baseline predictions establish a reference point for evaluating the incremental value of incorporating Fama-French factors.

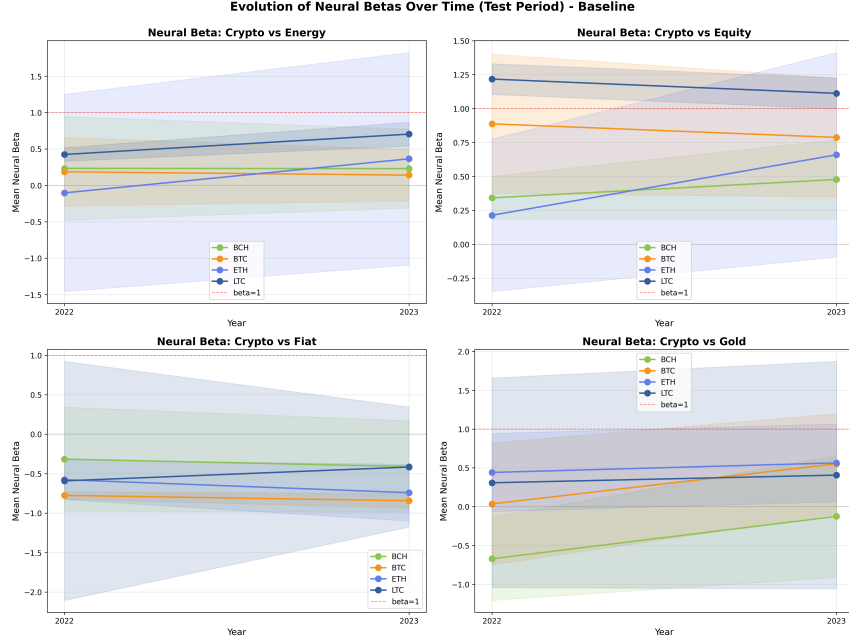


Figure 3: Beta Dynamics over time based on baseline model

Figure 4 shows that the incorporation of Fama-French factors into the neural architecture produces markedly different beta trajectories compared to the baseline model, suggesting that size, value, and momentum exposures contain meaningful information for cryptocurrency systematic risk. Most striking is the crypto-equity relationship, where the Fama-French enhanced model predicts declining betas for all assets, with BTC dropping from approximately 1.35 to -0.4 and ETH declining from 1.15 to near-zero. This contrasts sharply with the baseline model's stable predictions and may indicate that the factor-augmented network is capturing regime shifts in crypto-equity correlations that occurred during this period of market stress. In the crypto-energy space, the enhanced model predicts convergence of all four assets toward neutral exposure by 2023, with ETH exhibiting the most dramatic shift from -0.75 to +0.8. The crypto-fiat predictions show similar convergence patterns but with more compressed ranges, while the crypto-gold relationships display divergent trajectories with BCH predicted to transition from negative to strongly positive exposure (+1.3) and LTC moving in the opposite direction. The wider confidence bands in several panels suggest increased uncertainty in the factor-augmented predictions, which may reflect the complexity of capturing both traditional equity factors and cryptocurrency-specific dynamics simultaneously.

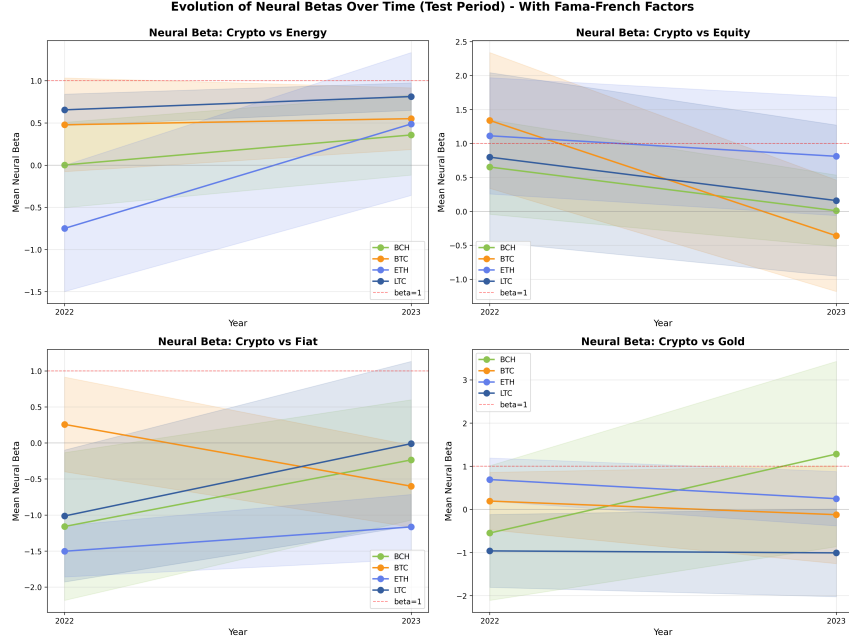


Figure 4: Beta Dynamics over time based on Fama-French enhanced model

3.4 Model Performance

Figure 5: The comparative performance analysis reveals nuanced effects of Fama-French factor augmentation on prediction accuracy across different cryptocurrency-asset pairs. The left panel shows that absolute RMSE values are highest for the ETH and LTC pairs against equity and fiat, ranging from 0.29 to 0.33, while BCH consistently exhibits the lowest prediction errors around 0.20-0.25 across most asset classes. Interestingly, the orange bars (Fama-French enhanced model) closely track the blue bars (baseline) for most pairs, with differences often imperceptible at this scale. The right panel quantifies these differences more precisely, revealing that the Fama-French factors provide modest improvements for several pairs, most notably ETH-energy (approximately 5% improvement) and LTC-equity (approximately 3% improvement), shown in green. However, the model also exhibits degradation for certain pairs, particularly LTC-gold and LTC-fiat (approximately 6% and 3% worse respectively), indicated in red. The mixed results suggest that while traditional equity risk factors capture meaningful variation in some crypto-asset relationships—particularly for energy and equity exposures—they may introduce noise or model complexity that hinders prediction accuracy for more idiosyncratic relationships like crypto-gold pairs. This heterogeneity across asset pairs indicates that the value of factor augmentation is context-dependent and may benefit from a more selective application based on the specific cryptocurrency and benchmark asset under consideration.

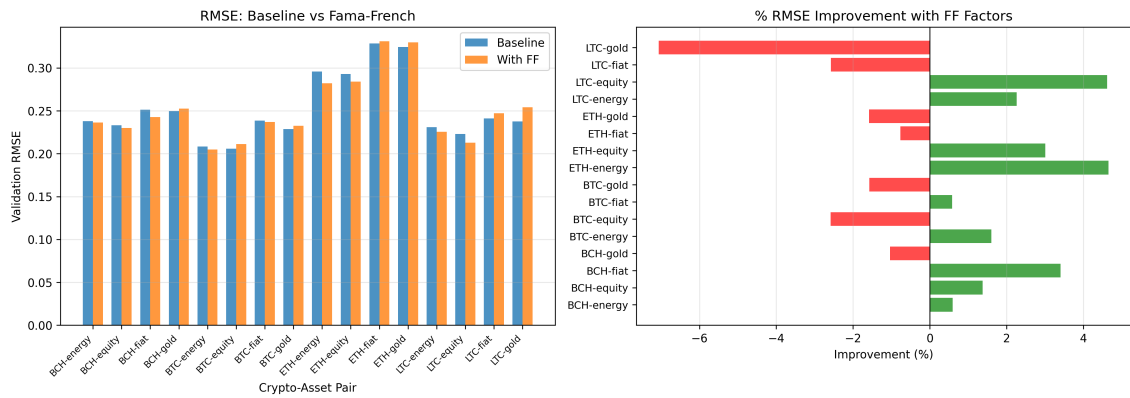


Figure 5: Model Performance Comparison Across Different Asset Pairings